

REVISION 2026

Basic Surveying Lab

Diploma in Civil Engineering

Code: 2019

Credits: 1.5

Semester: 2

Course Outcomes (CO)

- **CO1:** Perform linear and angular measurements using chain and compass. (Apply)
- **CO2:** Determine level differences using simple and differential levelling. (Apply)
- **CO3:** Plot longitudinal profiles and prepare contour maps. (Apply)
- **CO4:** Measure horizontal angles and bearings using a theodolite. (Apply)

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module 1: Chain & Compass Survey

Expt 1: Equipment Introduction

Demonstration of basic survey equipment: Chain, Cross-staff, Plane table, and Compass.

Expt 3: Compass Survey

Determine inaccessible distances between points using compass and tape measurements.

Experiments, assignments and projects are provided to help students understand the concepts and applications of surveying.

Module 2: Levelling

Expt 4: Simple Levelling

Determine level difference between two points using a Dumpy level.

Expt 5: Rise and Fall Method

Determine reduced levels (RL) by the Rise and Fall method for a series of points.

Experiments, assignments and projects are provided to help students understand the concepts and applications of levelling.

Module 3: Profile & Contouring

Expt 7: Profile Levelling

Plot longitudinal profiles and cross sections for the alignment of a road.

Expt 9: Contouring

Take spot levels and prepare a contour map for a given area.

[illegible]

Expt 10: Repetition Method

Measure horizontal angles with high precision using the repetition method with a theodolite.

Expt 12: Prolonging a Line

Prolong a straight line and measure the bearing of a line using a theodolite.

[illegible]

1. Determine the level difference between two given points using the Height of Instrument (HI) method.
2. Measure the horizontal angle between two objects using the repetition method with a theodolite.

Source Declaration

Curriculum Scheme: Revision 2026

Subject Name: Basic Surveying Lab

Subject Code: 2019

Official Source URL: [SITTTR Link](#)

Source Verification Date: 2026-08-19

This handbook was generated only from the identified scheme-specific SITTTR source.
Content from other curriculum revisions was not merged.